Abhinav Chakraborty

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Department of Statistics, Columbia University

Curriculum Vitae

Academic Positions

Columbia University *New York, NY*
Founder's Postdoctoral Fellow, Department of Statistics 2025–present

Education

University of Pennsylvania *Philadelphia, PA*
Ph.D. in Statistics, The Wharton School 2020–2025
Advisor: T. Tony Cai

Indian Statistical Institute *Kolkata, India*
M.Stat., specialization in Probability 2018–2020
Ranked first; **Best master's dissertation**; First Division with Distinction

Indian Statistical Institute *Kolkata, India*
B.Stat. (Honours) 2015–2018
Ranked first; First Division with Distinction

Publications

Author notation: $(\alpha\beta)$ Alphabetical ordering. * Equal contribution.

1. $(\alpha\beta)$ T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Optimal federated learning for functional mean estimation under heterogeneous privacy constraints. In press, *Journal of the Royal Statistical Society: Series B*, 2026. [paper]

2. $(\alpha\beta)$ T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Optimal federated learning for nonparametric regression with heterogeneous distributed differential privacy constraints. In press, *Journal of the American Statistical Association*, 2026. [paper]

3. **Abhinav Chakraborty**, Sayak Chatterjee, and Sagnik Nandy. PriME: Privacy-aware membership profile estimation in networks. In press, *SIAM Journal on Mathematics of Data Science*, 2026. [paper]

4. $(\alpha\beta)$ T. Tony Cai, **Abhinav Chakraborty**, and Yichen Wang. Optimal differentially private ranking from pairwise comparisons. *Journal of the American Statistical Association*, 2026. [paper]

5. $(\alpha\beta)$ Arnab Auddy, T. Tony Cai, and **Abhinav Chakraborty**. Minimax and adaptive transfer learning for nonparametric classification under distributed differential privacy constraints. *Journal of the Royal Statistical Society: Series B*, 88(3):903–929, 2026. [paper]

6. **Abhinav Chakraborty**, Arnab Auddy, and T. Tony Cai. When data can't meet: Estimating correlation across privacy barriers. *Advances in Neural Information Processing Systems*, 38, 2025. (Spotlight) [paper]

7. Ziang Niu*, **Abhinav Chakraborty***, Oliver Dukes, and Eugene Katsevich. Reconciling model-X and doubly robust approaches to conditional independence testing. *The Annals of Statistics*, 52(3):895–921, 2024. [paper]

8. **Abhinav Chakraborty***, Anirban Chatterjee*, and Abhinandan Dalal*. PrIsing: Privacy-preserving peer effect estimation via Ising model. *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, PMLR 238:2692–2700, 2024. [paper]

Preprints

1. Lucas Kania*, **Abhinav Chakraborty***, Edward Kennedy, Larry Wasserman, and Sivaraman Balakrishnan. Optimal inference with black-box predictions. arXiv preprint arXiv:2608.10155, 2026. [paper]

2. **Abhinav Chakraborty** and Sagnik Nandy. Transfer learning in high-dimensional clustering: Minimax thresholds and applications in single-cell data. arXiv preprint arXiv:2607.25031, 2026. [paper]

3. **Abhinav Chakraborty** and Subha Maity. The statistical cost of adaptation in multi-source transfer learning. arXiv preprint arXiv:2605.09471, 2026. [paper]

4. **Abhinav Chakraborty***, Yuetian Luo*, and Rina Foygel Barber. Stability and accuracy trade-offs in statistical estimation. arXiv preprint arXiv:2601.11701, 2026. Under major revision at *IEEE Transactions on Information Theory*. [paper]
5. **Abhinav Chakraborty***, Junu Lee*, and Eugene Katsevich. Comparing three learn-then-test paradigms in a multivariate normal means problem. arXiv preprint arXiv:2601.07764, 2026. [paper]
6. $(\alpha\beta)$ T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. The cost of adaptation under differential privacy: Optimal adaptive federated density estimation. arXiv preprint arXiv:2512.14337, 2025. [paper]
7. Sayak Chatterjee, Anirban Chatterjee, **Abhinav Chakraborty**, and Bhaswar B. Bhattacharya. Asymptotic normality of subgraph counts in sparse inhomogeneous random graphs. arXiv preprint arXiv:2512.12937, 2025. [paper]
8. **Abhinav Chakraborty***, Jeffrey Zhang*, and Eugene Katsevich. Doubly robust and computationally efficient high-dimensional variable selection. arXiv preprint arXiv:2409.09512, 2024. [paper]
9. $(\alpha\beta)$ T. Tony Cai, **Abhinav Chakraborty**, and Lasse Vuursteen. Federated nonparametric hypothesis testing with differential privacy constraints: Optimal rates and adaptive tests. arXiv preprint arXiv:2406.06749, 2024. R&R at *JMLR*. [paper]
10. **Abhinav Chakraborty**, Soumendu Sundar Mukherjee, and Arijit Chakrabarti. High dimensional PCA: A new model selection criterion. arXiv preprint arXiv:2011.04470, 2020. [paper]

Awards and Honors

IMS Travel Award 2025
Frontiers in Statistical Machine Learning Workshop.

Travel Award, Optimization and Statistical Learning Workshop 2025
Columbia University.

Lawrence D. Brown Best Student Paper Award 2024
The Wharton School, University of Pennsylvania.

George James Term Award 2022, 2024
Institute of Mathematical Statistics Annual Meeting.

ISI Alumni Association J. K. Ghosh Memorial Gold Medal 2020
Top scorer in the M.Stat. program.

P. C. Mahalanobis Gold Medal 2020
Best master's dissertation; outstanding M.Stat. student.

ISI Alumni Association Mrs. M. R. Iyer Memorial Gold Medal 2018
Top scorer in the B.Stat. program.

Nikhilesh Bhattacharya Gold Medal 2018
Highest marks in Statistics in the B.Stat. (Hons.) program.

INSPIRE Scholarship 2015
Awarded to students in the top 1% of the Indian School Certificate Examination.

Indian National Mathematics Olympiad qualifier 2014
Top-30 candidate in the Regional Mathematics Olympiad, 2013.

Talks

MCP Conference, Philadelphia, PA 2025
Invited talk: Learn-then-test procedures.

IISA Conference, Lincoln, NE 2025
Invited talk: Federated transfer learning.

IMS Annual Meeting, Bochum, Germany 2024
Federated nonparametric regression.

Ph.D. Student Seminar, University of Pennsylvania 2023
Optimal distributed private estimators.

IMS Annual Meeting, London, UK 2022
High dimensional PCA.

P. C. Mahalanobis Gold Medal Presentation, Indian Statistical Institute 2020
Model selection in high dimensional PCA.

D. Basu Memorial Presentation, Indian Statistical Institute 2018
MCMC methods in Bayesian inference.

Teaching and Mentoring

Teaching Assistant, The Wharton School, University of Pennsylvania 2021–2024
Introduction to Probability (STAT 4300); Probability Theory (MATH 6480); Mathematical Statistics (STAT 971); Applied Econometrics (STAT 520).

Ph.D. Peers Program, The Wharton School, University of Pennsylvania 2023
Mentored an incoming Wharton Ph.D. student in navigating the program and transitioning to Penn.

Professional Service

Journal reviewer: *The Annals of Statistics*; *Journal of the American Statistical Association (JASA)*; *IEEE Transactions on Information Theory*; *Journal of the Royal Statistical Society: Series B (JRSSB)*; and *Biometrika*.